

Manisha Kaler

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EDUCATION

University of Massachusetts

Jan 2026 - Dec 2027 (Expected)

Masters of Science in Computer Science

3.9/4.0

Coursework: Machine Learning, Performance Engineering, Data Science Fundamentals

Indian Institute of Technology

Aug 2017 - May 2021

Bachelors of Technology in Computer Science and Engineering

Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Networks, Operating Systems, Computer Organization and Architecture, Algorithms Design and Analysis, Theory of Computation, Computer Graphics

WORK EXPERIENCE

Arun Data Processing

Software Engineer 2

Feb 2023 - Dec 2025

- Built and shipped a **Go-based PDF generation microservice** for the Income Tax Department of India's production tax filing workflow, supporting **80M+ taxpayers**.
- Engineered **schema-driven rendering logic** that mapped nested JSON structures to reusable **Go templates** for tax forms, acknowledgements, and verification PDFs.
- Benchmarked third-party **HTML-to-PDF renderers** and selected a compatible solution to improve PDF generation reliability for legacy **RHEL Linux** infrastructure.
- Built a **schema-driven code generation pipeline** that generated Angular HTML, TypeScript, and service-layer code, reducing new workflow delivery time from **4 months to 3 weeks**.
- Developed **30+ Angular components** and shared modules for validation, localization, and **WCAG-aligned accessibility** across online and offline filing flows.
- Integrated **20+ client REST APIs** end-to-end and packaged the Angular utility into cross-platform desktop executables using **Wails**.

VISA

Software Engineer

Jul 2021 - Jan 2023

- Optimized ScalableRun, a **global script execution platform** managing **10,000+ servers**, improving platform reliability through **real-time validations** and **asynchronous processing**.
- Designed event-driven pub-sub systems using **message queues** to monitor **Ansible Tower** automation, surfacing execution metrics in **Grafana dashboards** to reduce **MTTR by 30%**.
- Implemented the frontend for **Ansible group management**, enabling users to organize nested server groups for bulk script execution and improving deployment speed by **20%**.
- Built an **IAM-based access validation layer** to enforce user permissions across server groups, ensuring secure script executions across global **Cloud PaaS** infrastructure.
- Built **unit and integration tests** for Spring Boot services using **JUnit**, **Mockito**, and Spring test utilities, achieving **95% code coverage**.

VISA

Software Engineer Intern

May 2020 - Jul 2020

- Developed a full-stack web application using **Java** and **Visa Developer APIs** to provide **geo-targeted offer recommendations** for small merchants.
- Prototyped a functional **payroll API** solution in Java, digitizing merchant payment workflows for a global case challenge.

PROJECT EXPERIENCE

Lecture Notes AI Assistant | *LLM-based Study Tool*

UMass Amherst

- Built an **LLM-powered** study assistant using **Streamlit** and **Gemini 2.5 Flash-Lite** to summarize transcripts, answer grounded questions, and generate study notes.
- Implemented a structured storage layer with **SQLAlchemy** and **SQLite** to persist transcripts and support retrieval grounded responses.

Cache-Optimized AVX Matrix Multiplication Kernel | *C++, AVX, OpenMP, Linux perf*

UMass Amherst

- Optimized **C++ matrix multiplication** with loop reordering, cache blocking, **AVX vectorization**, register-aware kernels, and **OpenMP**; achieved **4.48x** single-thread and **2.44x** 8-thread speedup on large workloads.
- Used **Linux perf** and assembly inspection to analyze performance, reducing **L1 cache miss rate** from **72.47%** to **6.56%**, then to **1.25%** with a cache-aware tiled kernel.

TECHNICAL SKILLS

Languages: Go, Java, TypeScript, JavaScript, C/C++, SQL, HTML/CSS, Python

Frontend: Angular, React, Next.js, Bootstrap, Wails, Responsive Design

Backend & APIs: Spring Boot, Gin, RESTful APIs, Microservices

AI & LLM: Prompt Engineering, RAG Pipelines, Gemini API, Grounding

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda), Docker, Jenkins, Ansible Tower, Linux, Git

Data & Tools: PostgreSQL, MySQL, MongoDB, Hazelcast, SQLite, SQLAlchemy, Grafana, perf

Testing & Process: Unit Testing, Integration Testing, Agile/Scrum